

Fitted curve results

Roallos, Gene Lowelle | Four-parameter logistic model

Fitted parameters

Parameter	Estimate	Standard error	t-statistic	p-value
c (lower asymptote)	14.378 m	0.038	379.31	0.00e+00
a (amplitude)	5.514 m	0.050	110.41	0.00e+00
k (growth rate)	0.855 1/h	0.062	13.69	0.00e+00
t0 (inflection time)	30.16 h	0.10	308.25	0.00e+00

Fit statistics and integration

SSE	42.412537 m ²	R ²	0.9791
s	0.3864 m	Degrees of freedom	284
Area under fitted curve	1260.9239 m·h	Quadrature error	4.58e-07 m·h

Reading of the residuals

The residuals are centred on zero throughout; there is no systematic vertical bias. Long runs of the same sign are present, which means the shape of the logistic model does not fully capture the curvature of the data (the model form is inadequate). The residual spread does not increase markedly with stage. The largest residual is 1.278 m, which is 127.8 times the logger's 1 cm resolution; the discrepancy is therefore larger than measurement noise.